



As highlighted in the image, screen tearing can sometimes totally ruin the experience

Most games and other video intensive applications implement something called VSync which basically limits the refresh rate of the GPU to a constant value (say 40fps from 60fps) once a discontinuity occurs, so that if any frame misses it's designated placement on the screen's refresh cycle, the GPU holds back sending any new frames until the next refresh. All's well? Not quite.

This forceful limiting of the refresh rate causes lag and stutters during high-CPU loads, which is almost 100% of the time when the gamer actually needs it to NOT lag and stutter. So most gamers willingly disable VSync and chose to live with the tearing rather than the lag. Which is understandable in it's own way. But someone had to do something about such a irritating problem in the otherwise wonderful experience of gaming. Enter NVIDIA.

GSync - Frame by Frame

When video games were suffering, how can companies like NVIDIA and AMD stay out of the battle? NVIDIA's proprietary G-Sync solution to the sync problem is being advocated by them since the last few years. This piece of technology puts the reigns of the monitor's frame rate into the GPU's hands. This is based on NVIDIA's previous work on Adaptive VSync where